

4. In addition to the ABO blood group system in humans there is another system called MN. It is caused by codominant alleles, M and N which code for proteins in the membrane of red blood cells.

- (a) (i) The Hardy-Weinberg principle states that allele frequencies in a population remain constant from generation to generation providing certain conditions exist. State **three** of these conditions. [3]

1.

2.

3.

- (ii) The frequency of allele N in Germany is 45%.
Calculate the following given that:

p = frequency of allele M,

q = frequency of allele N,

$p + q = 1$

$p^2 + 2pq + q^2 = 1$

- I. The % frequency of allele M; [1]

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- II. The number of individuals in a population of 10 000 with blood group M; [1]

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- III. The number of individuals in a population of 10 000 with blood group N; [1]

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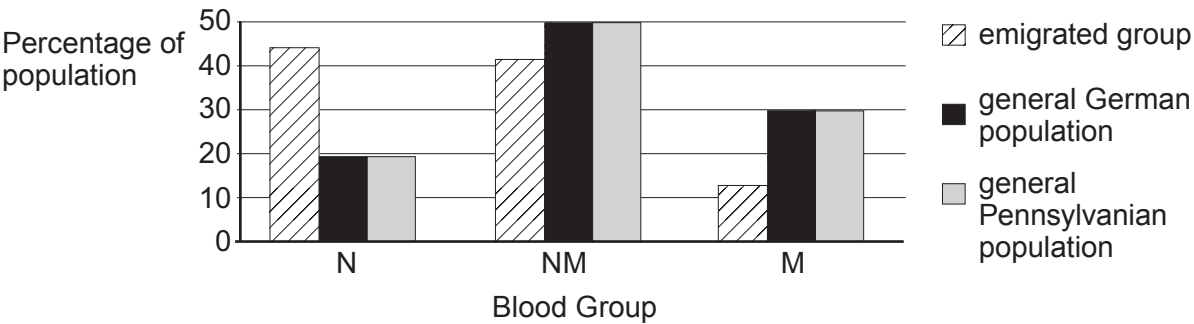
- IV. The number of individuals in a population of 10 000 with blood group MN. [1]

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(b) During the eighteenth century, some groups of people emigrated from Germany to America. Since then, these people have married almost exclusively within their own communities.
In 1950, a blood group analysis of 200 members from one of these groups in Pennsylvania (a state in America) was undertaken. Their blood groups were compared with current German and American populations. Blood group analyses produced the following results:



Explain why the emigrated population shows allele frequencies which are uncharacteristic of either the American or German populations. [3]

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